POLITECNICO
MILANO 1863

Radar Imaging

Homework #2

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1. Definition of the Project

In this project we want to use the Time Domain Back Projection (TDBP) algorithm to receive an image of the environment. In general, these images were obtained using Unmanned Aerial Vehicles (UAVs). This method combines radar return signals and focuses them on a grid of image points to convert the received reflections into an image. The TDBP algorithm is used in radar image reconstruction and is used to create high-resolution images of targets.

2. Range Resolution Assessment

First, we calculate range resolution (ΔR) with knowing the bandwidth value according to the below equation.

$$\Delta R \cong c/(2B)$$

Where c is a speed of light and B is a bandwidth.

```
Command Window
>> Range_Resolution_with_Bandwidth
Range Resolution: 0.38 meters
```

Fig. 1. Result of the range resolution

Then, we calculate range resolution (ΔR) without knowing the value of bandwidth. First, we calculate the sampling frequency, and then perform the FFT for each radar pulse. Then, we estimate bandwidth and compute range resolution is equal to 0.42.

```
Command Window
The computed bandwidth is: 358478484.25 Hz
Range Resolution Estimate: 0.42 meters
```

Fig. 2. Result of the range resolution using an energy threshold

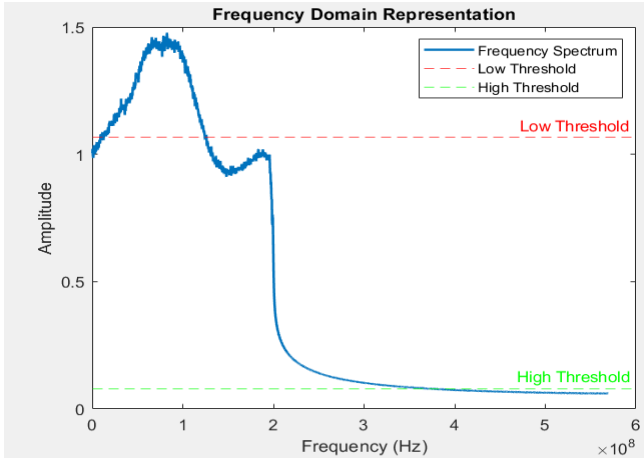

Fig. 3. Frequency Domain Representation with Threshold Line

3. Time Domain Back Projection

In this part of the task, we want to create an SAR image using the TDBP algorithm. We use this method and obtain high-resolution images as possible.

Fig. 4. Optical Image

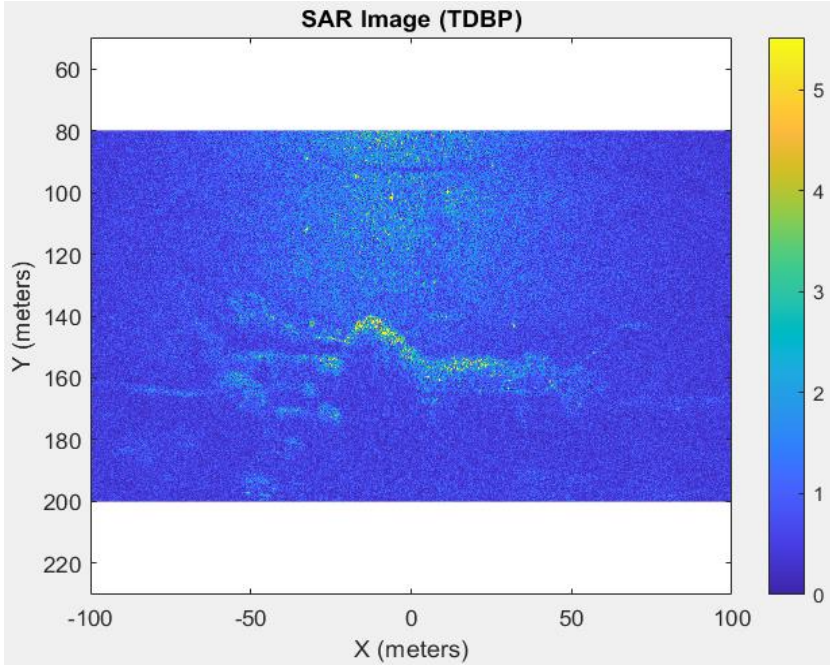

Fig. 5. Radar Image

By comparing the two images above, we find that the radar image is closest to the middle parts of the optical image, which indicates the efficiency of the algorithm used. The TDBP algorithm requires a lot of calculations and is time-consuming. So, we calculated the time required to implement the algorithm in MATLAB and show it in the below figure.

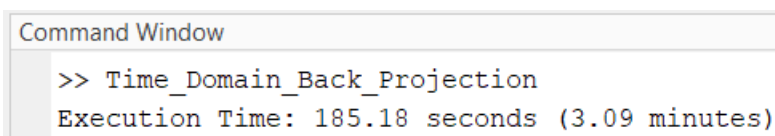

Fig. 6. Execution Time

4. Despeckling

In this part of the project, we want to decrease the speckle noise of the image. This noise makes it difficult to accurately identify image features and details. We use different sizes of moving averages and apply them to the image. The obtained image has less noise than the original image. Following, we show these images.

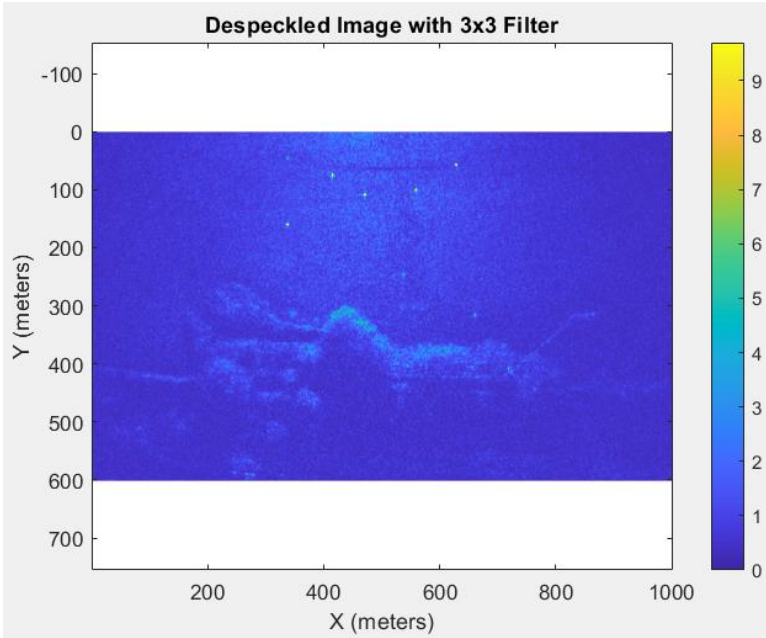

Fig. 7. Despeckled Images with 3*3 Filter

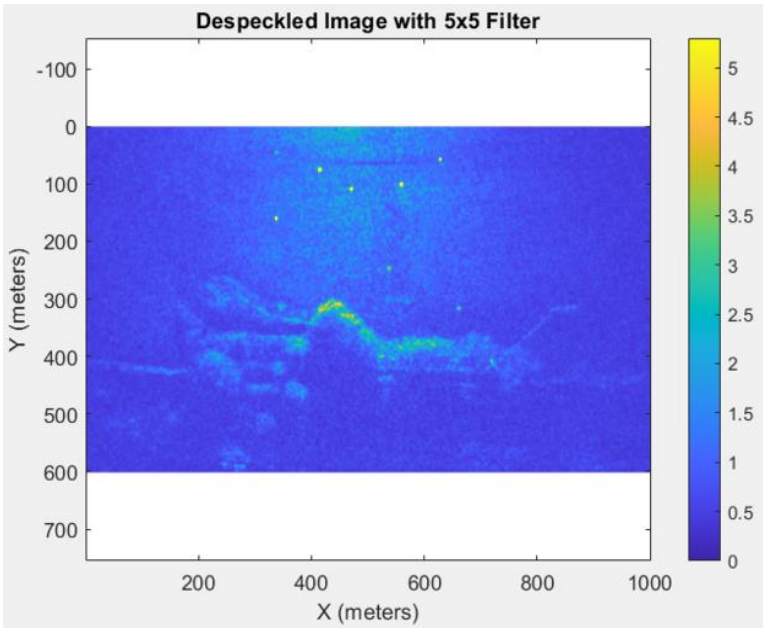

Fig. 8. Despeckled Images with 5*5 Filter

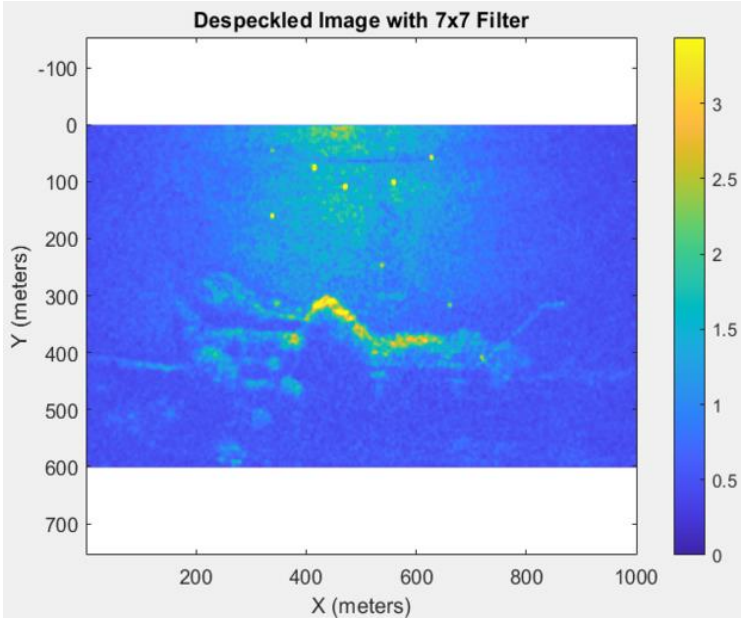

Fig. 9. Despeckled Images with 7*7 Filter

By comparing the images, we come to the conclusion that the larger the size of the applied filter, the lower the image noise, but some of the details of the image are lost. In contrast, a smaller filter preserves image detail better but has less effect on noise removal. Therefore, determining the appropriate filter to remove noise is one of the important features in the process of removing noise from images.

5. Corner Reflectors

Corner reflectors are very bright and distinct targets placed in the scene for calibration and appear clearly in images due to their strong reflection of radar waves. In this section we should calculate two types of resolution: 1- range resolution 2- azimuth resolution. We calculate range resolution in the Y- axis direction based on FFT. Also, we compute azimuth resolution based on the below equation.

$$Azimuth_Resolution = \frac{\lambda . r_P}{2 . L_s}$$

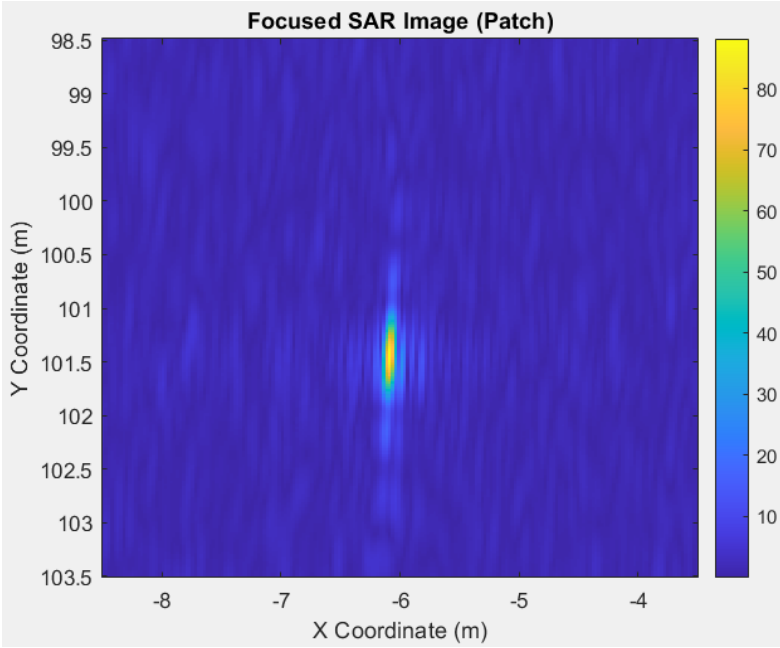

Fig. 10. Focused SAR image (Patch)

Fig.10. shows the intensity of reflections at different points in the patch.

```
Command Window
>> Corner__Reflectors
Azimuth Resolution: 0.3802 meters
Resolution in Y direction (FFT-based): 0.4762 meters
```

Fig. 11. Result of Azimuth Resolution and Resolution in Y Direction

6. The Effect of Trajectory Errors

In the last part of the task, we want to see the effect of noise on the SAR image by adding noise to the UAV trajectory. We want to compare the effect of this increase in noise to the UAV trajectory on the quality of the image. The more noise applied, the more details are lost from the image. Also, we calculated the time required for this part of the project. As you can see in the images below, the higher the standard deviation, the more distorted and blurrier the image becomes, and the more detail is lost. The most image distortion occurs when we apply standard deviation $\lambda/2$.

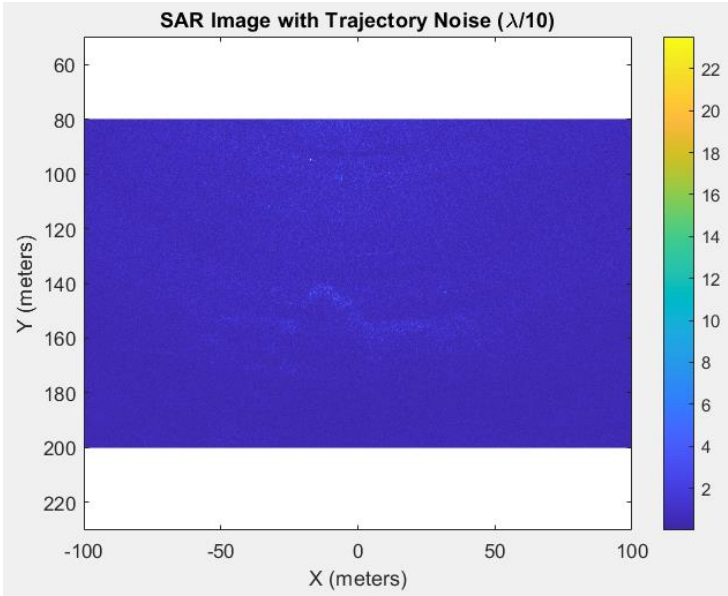

Fig. 12. SAR image with standard deviation $\lambda/10$

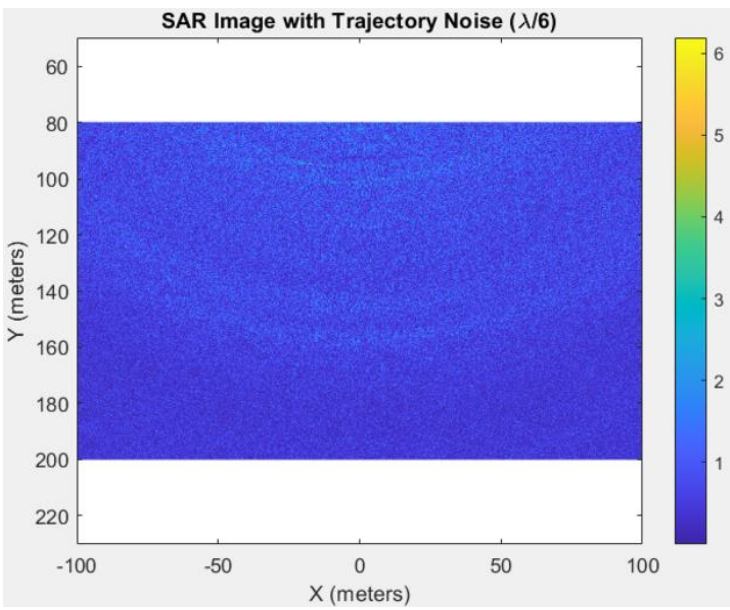

Fig. 13. SAR image with standard deviation $\lambda/6$

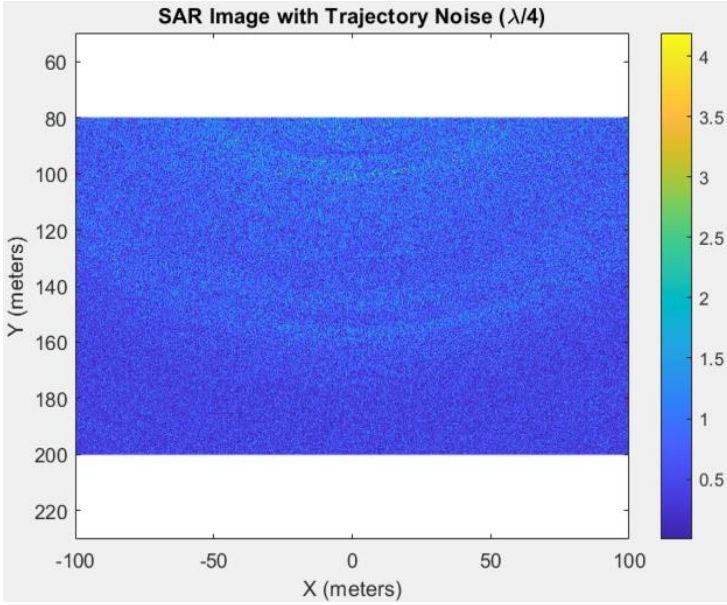

Fig. 14. SAR image with standard deviation $\lambda/4$

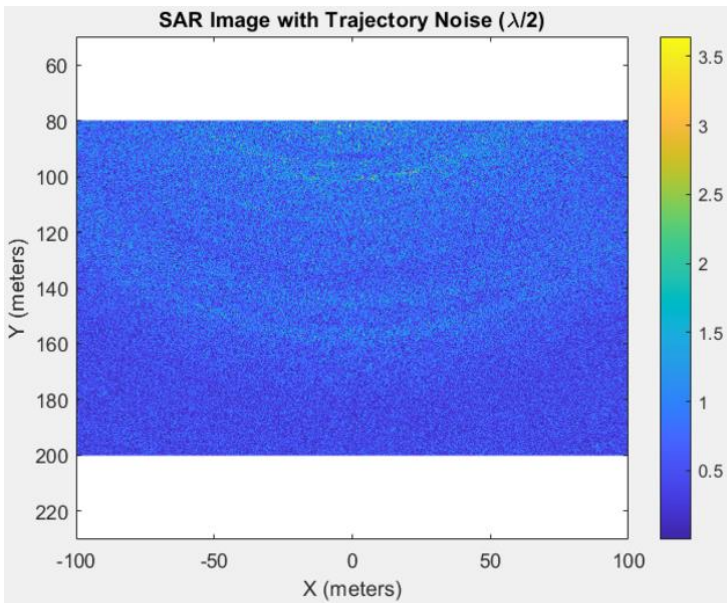

Fig. 15. SAR image with standard deviation $\lambda/2$

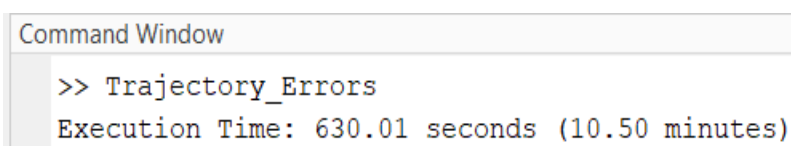

Fig. 16. Execution Time